

## Assignment - 02

### Ans to the q. no-01

Absolute Error: This is the raw difference between the measured value ( $n$ ) and the actual value ( $n_0$ ). It's expressed as

$$\Delta n = |n - n_0|$$

Relative Error: This considers the significance of the error relative to the actual value. It's calculated by dividing the absolute error by the actual value and expressed as a percentage:

$$\text{Relative Error} = (|n - n_0| / n_0) \times 100\%$$

Error Percentagile (from relative Error):

1. Get the absolute value from relative error
2. Multiply by 100%.
3. Calculate the absolute value of the

relative error:  $|(x - x_0)/x_0| \times 100\%$ .

This is error percentile.

Ans to the q: no-02

To find the root of the equation

$$y = 5\cos^3 x - e^{4x}$$

using bisection method, we will follow these steps-

1. Chose the interval  $[a, b]$  such that  $f(a)$  and  $f(b)$  have opposite sign

2. Calculate the midpoint  $c$  of the interval

$$c = \frac{a+b}{2}$$

3. Evaluate  $y(c)$

4. Determine the new interval based on the sign of  $f(c)$

(i)  $f(a) \times f(c) < 0$  then set  $b = c$

(ii)  $f(a) \times f(c) > 0$  then set  $a = c$

(iii)  $f(a) \times f(c) = 0$  then root equals to  $c$

Determine the equation.

5. Repeat the step until count reach at 10.

choosing the interval value.

$$a=0, \quad b=1$$

$$f(0) = 5 \times (\cos(0))^3 - e^{4 \times 0} = 4$$

$$f(1) = 5 \times (\cos(1))^3 - e^{4 \times 1} = -53.98$$

| Iteration | a                           | b      | $c = \frac{a+b}{2}$ | $f(c)$  |
|-----------|-----------------------------|--------|---------------------|---------|
| 1         | 0                           | 1      | 0.5                 | -4.0097 |
| 2         | <del>0.45</del>             | 0.5    | 0.25                | 1.8298  |
| 3         | 0.25                        | 0.5    | 0.375               | -0.4533 |
| 4         | <del>0.25</del> 0.25        | 0.375  | 0.3125              | 0.8178  |
| 5         | <del>0.3438</del><br>0.3125 | 0.375  | 0.3438              | 0.2177  |
| 6         | 0.3438                      | 0.375  | 0.3594              | -0.1085 |
| 7         | 0.3438                      | 0.3594 | 0.3516              | 0.0569  |
| 8         | 0.3516                      | 0.3594 | 0.3555              | -0.0252 |
| 9         | 0.3516                      | 0.3555 | 0.3535              | 0.0166  |
| 10        | 0.3535                      | 0.3555 | 0.3545              | -0.0057 |

The root is approximately  $\approx 0.3545$

### Ans to the q. no-03

To find the root of equation  $y = 5\cos^3 x - e^{4x}$  using the Newton-Raphson method, we need to follow these steps.

Newton-Raphson Method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Step-1: Define the function and its derivative function.

$$f(x) = 5\cos^3 x - e^{4x}$$

$$f'(x) = -15\cos^2 x \cdot \sin x - 4e^{4x}$$

Step-2: Chose an initial guess  $x_0$ . Choosing a good initial guess is important for the convergence of the method. Factors to consider for the initial guess included.

Iteration - 1:

let chose  $x_0 = 0$

$$f(0) = 5\cos^3(0) - e^0 = 4$$

$$f'(0) = -15 \cos^2(0) \cdot \sin(0) - 4e^0 = -4$$

$$x_1 = x_0 - \frac{f(x)}{f'(x)} = 0 - \frac{4}{-4} = 1$$

Iteration-2:

$$f(1) = 5 \cos^3(1) - e^4 = -53.8095$$

$$f'(1) = 5 \cdot \cos^2(1) \cdot \sin(1) - 4e^4 = -222.077$$

$$x_2 = 1 - \frac{-53.8095}{-222.077} = 0.7576$$

Iteration-3:

$$f(0.7576) = 5 \cos^3(0.7576) - e^{4 \times 0.7576} = -18.7884$$

$$f'(0.7576) = -15 \cos^2(0.7576) \cdot \sin(0.7576) - 4e^{4 \times 0.7576} \\ = -88.2628$$

$$x_3 = 0.7576 - \frac{-18.7884}{-88.2628}$$

$$= 0.544$$

Iteration-4:

$$f(0.544) = 5 \cos^3(0.544) - e^{4 \times 0.544} = -5.708$$

$$f'(0.544) = 5 \cos^2(0.544) \cdot \sin(0.544) - 4e^{4 \times 0.544} \\ = -41.032$$

$$x_4 = 0.544 - \frac{-5.70869}{-41.0327} = 0.40566$$

using Newton-Raphson method, the approximate root of the equation  $y = 5\cos^3 x - e^{4x}$  is around  $x \approx 0.4056$  after 4 iteration

Factors for initial Guess

- Analyzing the function at key points shows a sign change
- Given that  $\cos(x)$  oscillates, we consider values where  $e^{4x}$  grows quickly.

## Ans to the q. no-04

A tabulated function is a function where values are pre-calculated and stored in a table.

### Maxima and Minima identification Algorithm

1. Input: Tabulated function values  $T$  as a list.

2. Initialize: Create maxima or minima lists.

3. Iterate:

For  $i$  from 1 to  $n-2$

if  $T[i] > T[i-1]$  and  $T[i] > T[i+1]$ , add  $i$  to maxima

if  $T[i] < T[i-1]$  and  $T[i] < T[i+1]$ , add  $i$  to minima

4. Boundary check:

check  $T[0]$  and  $T[n-1]$  for maxima/minima

5. Output: Return the list of maxima and minima  
indices.

Example:

For  $T = [1, 3, 2, 5, 4, 6, 3, 1]$ :

→ Maxima: Indices  $[1, 3, 5]$

→ Minima: Indices  $[2, 4, 6]$

Ans to the q. no-0.5

Local Maxima and Global Maxima:

Local Maxima: A point  $x_i$  in the function  $f(x)$  is a local maximum if  $f(x_i) \geq f(x)$  for all  $x$  in some neighbourhood around  $x_i$ .

Global Maxima: A point  $x^*$  is a global maximum if  $f(x^*) > f(x)$  for all  $x$  in the domain of the function.

Justification for "Newton's divided difference method always finds the global maxima"

Newton divided difference method always not finds the global maxima. It is used for polynomial interpolation, not optimization. It builds polynomials that pass through given data points. The statement that it "always finds the global maxima" is incorrect.

Ans to the q: no-06

Convergence vs. Divergence in Root Finding Algorithms:

Convergence: An algorithm converges if it approaches a solution (root) with each iteration, meaning the approximations get closer to the actual root.

Divergence: An algorithm diverges if it fails to approach a solution, with approximation moving away from the root or not stabilizing.

### Opening Methods and Divergence.

Opening methods can be more prone to divergence due to:

1. Poor initial Guess
2. Function Behaviour
3. Sensitivity
4. Iteration stability.

In summary, while opening methods can be effective, they require careful selection of initial points.

### Ans to the q: no-07

To integrate the function  $f(x) = \sin x - \cos x + x^5 - x^4$  over the interval  $a = -5$  to  $b = 5$  using trapezoidal rule and identify the error, we can follow the steps-

Trapezoidal rule: Approximate the integral  $\int_a^b f(x) \cdot dx$  by dividing the interval  $[a, b]$  into  $n$  sub-intervals of equal width  $h = \frac{b-a}{n}$  and summing up to the areas of the trapezoids under the curve, the formula is-

$$\int_a^b f(x) dx \cong \frac{h}{2} \left[ f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$$

1. chose the number of subintervals  $n$ . Let's for use  $n=5$  from simplicity.

2. calculate the width of each subintervals,  
 $h = \frac{5-(-5)}{5} = 2$

3. Calculate  $x_i$  values  $x_0 = -5$ ,  $x_1 = -3$ ,  
 $x_2 = -1$ ,  $x_3 = 1$ ,  $x_4 = 3$ ,  $x_5 = 5$

4. Calculate  $f(x)$  at those points

$$f(x) = \sin^2 x - \cos^5 x + x^5 - x^4$$

$$f(-5) = -3749.08$$

$$f(3) = 162.971$$

$$f(-3) = -323.029$$

$$f(5) = 2500.92$$

$$f(-1) = -1.33797$$

$$f(1) = 0.662028$$

Applying the trapezoidal formula

$$\int_{-5}^5 f(x) dx = \frac{2}{2} \left[ f(-5) + f(5) + 2 \left[ f(-3) + f(-1) + f(1) + f(3) \right] \right]$$

$$= -278.231885$$

Error: The error  $E_T$  of the trapezoidal

rule given by  $E_T = \frac{(b-a)^3}{12n^2} f''(\xi)$

## Ans to the q: no-08

### Derivation of Simpson's 1/3 Rule:

Simpson's 1/3 rule approximates the integral of a function using a quadratic polynomial.

Here is a simple derivation:

1. Interval Setup: Let  $[a, b]$  be the interval and divide it into two equal parts:

$$h = \frac{b-a}{2}$$

points are  $a, m = a+h, b$ .

2. Quadratic Polynomial: Assume  $f(x)$  is approximated by a quadratic  $p(x)$  through points  $(a, f(a)), (m, f(m)), (b, f(b))$ .

3. Integral calculation:

$$\int_a^b p(x) dx = \frac{h}{3} [f(a) + 4f(m) + f(b)]$$

#### 4. Final Rule:

$$\int_a^b f(u) du \approx \frac{h}{3} [f(a) + 4f(m) + f(b)]$$

The formula provides an accurate approximation of the integral when the function can be represented well by a quadratic over the interval.

Ans to the q. no. 09

#### Comparative Error of Simpson's 1/3 and 3/8 Rules:

Error Formulas:

##### 1. Simpson's 1/3 Rule:

$$E_{1/3} \leq \frac{(b-a)^5}{180n^4} \max |f^{(4)}(u)|$$

## 2. Simpson's 3/8 Rule:

$$E_{3/8} \leq \frac{(b-a)^5}{80n^4} \max |f^{(4)}(x)|$$

### Comparisons

→ The 3/8 rule generally has smaller error than 1/3 rule for the same number of intervals.

→ Use 1/3 rule for function approximation well by quadratics; use the 3/8 rule for functions with more curvature.

### Example

Function:  $f(x) = e^x$  over  $[0, 1]$

1. Exact Integral:

$$\int_0^1 e^x dx = e - 1 \approx 1.71828$$

2. Simpson's 1/3 Rule (with  $n=2$ ):

points:  $f(0)=1, f(0.5) \approx 1.64872$

$$f(1) = 2.71828$$

$$\text{Approximation} \approx 1.71828$$

3. Simpson's 3/8 Rule (with  $n=3$ ):

points:  $f(0)=1, f(0.3333) \approx 1.39561$

~~$f(1) \approx$~~   $f(0.6667) \approx 1.94773$

$$f(1) \approx 2.71828$$

$$\text{Approximation} \approx 1.71828$$

Ans to the q. no-10

Integral to Approximate

$$\int_0^6 (u^5 - 3u^4 + 5u) du$$

Step 1:

$$f(u) = u^5 - 3u^4 + 5u$$

Step 2:

Interval Division

interval  $[0, 6]$ , size  $h = \frac{6-0}{6} = 1$

Numbers  $u_0 = 0, u_1 = 1, u_2 = 2, u_3 = 3, u_4 = 4$

$u_5 = 5, u_6 = 6$

$$f(u_0) = f(0) = 0$$

$$f(u_4) = 276$$

$$f(u_1) = f(1) = 3$$

$$f(u_5) = 1275$$

$$f(u_2) = f(2) = -6$$

$$f(u_6) = 3918$$

$$f(u_3) = f(3) = 15$$

Simpson's 3/8 Rule

$$\int_0^6 (u^5 - u^4 + 5u) du = \frac{3h}{8} [f(u_0) + f(u_6) + 3(f(u_1) + f(u_2) + f(u_4) + f(u_5)) + 2f(u_3)]$$

$$= \frac{3 \cdot 1}{8} [0 + 3918 + 3(3 - 6 + 276 + 1275) + 2(15)]$$

$= 3222$

Exact value of the integral, we need to integrate function directly.

$$\int_0^6 (u^5 - 3u^4 + 5u) du = \left[ \frac{u^6}{6} - \frac{3u^5}{5} + \frac{5u^2}{2} \right]_0^6$$

$= 3200.4$

The approximation is close but higher than exact value.